A runnable check, not a reserve policy

Run the supplied Python 3 file offline. No packages, paid service, account or investor data are needed. The original fixed-seed model is retained and packaged for reuse, with path/statistic exports and four aligned distribution charts.

One 40 M commitment; all risky tranches at time 0, one terminal payment in year 10. Baseline excludes fees, carry, taxes, leverage, recycling, intermediate cash and interest. Outcome probabilities 55/25/12/7/1% map to 0/2/6/25/100 x; they are chosen, not fitted market data.

Strategy	Budget and follow-on rule
A	24 initial checks of 1; idle 16 returned at par.
B	40 initial checks of 1; no reserve or idle cash.
C	24 initial checks of 20/24 plus 20 follow-on into 6 existing companies.
D	40 initial checks of 0.5 plus 20 follow-on into 10 existing companies.

C/D select the top quarter by an observable score independent of payoff. Follow-on dollars pay twice the price, so buy half the terminal payoff per dollar. No hindsight, extra ownership or guaranteed rights are granted.

Analytic mean versus one fixed-seed sample

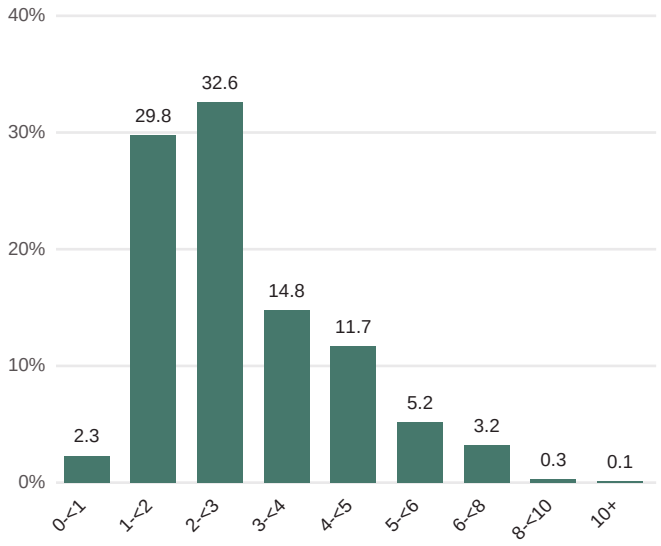
Strategy	Analytic EV	Sample mean	Median	P(>3 x)
A	2.782000	2.836950	2.450000	35.0%
B	3.970000	4.078200	3.750000	69.1%
C	2.977500	3.019000	2.541667	38.1%
D	2.977500	3.024725	2.693750	41.5%

C and D have the same analytic mean despite different sample means. This is not evidence that one reserve policy is superior. A deployed-only denominator gives 4.061583 x in this sample, not the commitment-based 2.836950 x.

Four aligned baseline distributions

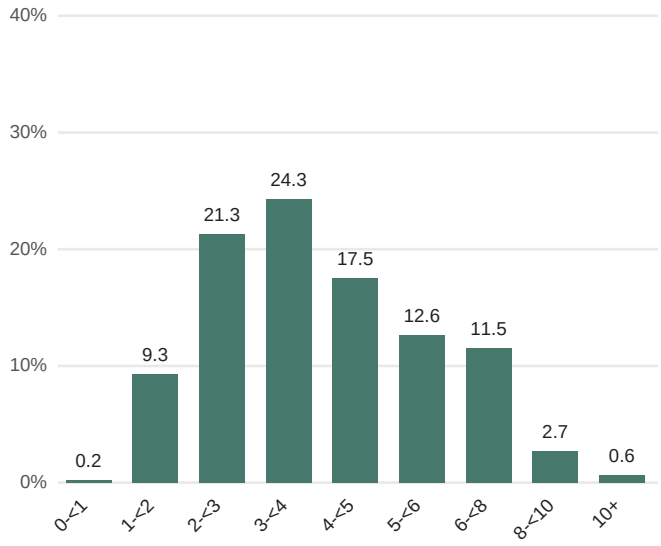
Bars show categorical probabilities from 1000 paired paths, not probability densities. Unequal-width bins have equal visual spacing. All panels use 0-40% vertical scales and the same bins.

Strategy A



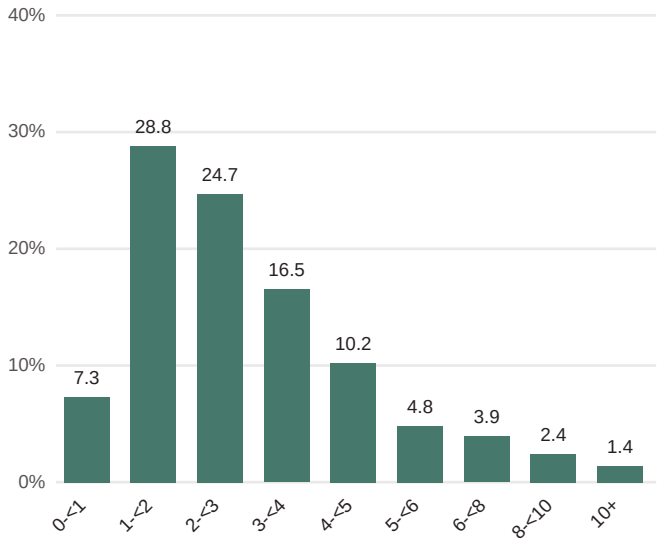
P(<1 x) 2.3% | mean MCSE 0.045233 x

Strategy B



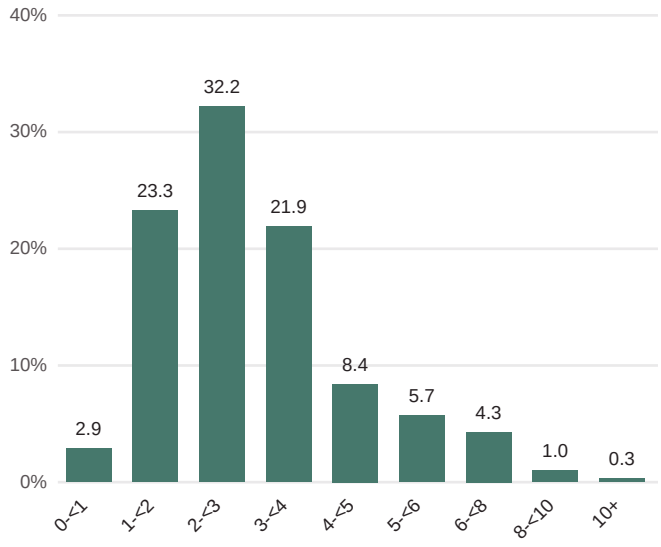
P(<1 x) 0.2% | mean MCSE 0.057490 x

Strategy C



P(<1 x) 7.3% | mean MCSE 0.064917 x

Strategy D



P(<1 x) 2.9% | mean MCSE 0.049600 x

Vertical units: percent of simulated paths. Horizontal units: terminal cash multiple of 40 M commitment. SVG and CSV contain exact bin counts; each panel sums to 1000.

Stress the mechanism, not just the mean

Scenario	A mean	B mean	C mean	D mean
baseline	2.836950	4.078200	3.019000	3.024725
tail 50	2.520700	3.538200	2.632542	2.630975
tail 0	2.204450	2.998200	2.246083	2.237225
common_shock	2.845094	4.045156	2.986432	2.959172
funded_fees_carry	2.241060	3.136149	2.368354	2.376130
hindsight_only	2.836950	4.078200	5.640000	5.692350

Tail 50 / tail 0

Replace only the 1% tail payoff, retaining its probability. Deal EV becomes 3.47 or 2.97 x; it does not automatically collapse to 1 x.

Compensated common shock

With 20% probability all risky payoffs are zero; otherwise multiply risky payoffs by 1.25. A idle 16 stays unchanged. This preserves each marginal EV analytically, not each sample mean. Baseline P(<1 x)2.3/0.2/7.3/2.9% becomes 20.8/19.8/23.1/20.7%.

Funded fees / profit-only carry

4 of 40 pays total fees; scale risky allocations AND idle cash to 90%, leaving 36 for the strategy. Apply 20% carry only on positive terminal profit above 40 paid-in. This is one stipulated waterfall, not legal terms, annual fees or a generic net-return rule.

Hindsight only

Rank realized future payoffs for C/D only as a nonimplementable diagnostic. Never call these results a deployable investment policy. Real score edge, access, dilution and follow-on pricing require separate evidence.

All scenarios share the specified outcome/score/shock uniforms, preserving a paired comparison. The random seed is a reproducibility choice, not evidence that the model describes real venture returns.

Reproduce, inspect and challenge

Run

python3 d162_model.py --output my-d162-output
The command creates the output folder and overwrites its fixed-named result files. Standard library only; no network requests or personal data.

Inspect

summary.json; scenario-statistics.csv; all-paths.csv with 1000 rows per scenario; baseline-bins.csv; baseline-distributions.svg. Reference outputs are included. Python random.Random(20261110):40 outcome draws,40 score draws,1 reserved shock draw each path, including baseline.

Control checks

24,000 independent path comparisons across six scenarios using integer-sixths allocations and exact rational shock/carry stages;48 strict-threshold comparisons;36 prior bin counts;12 baseline references and budget/tie/constant/boundary checks. Exact 3 x does not count as >3 x. Prior model core was reused, not newly devised.

Statistics

Nearest-rank percentiles use ceil(p*N)-1 in sorted values. SD is sample SD; mean MCSE=SD/sqrt 1000. Probability MCSE=sqrt(p*(1-p)/1000), reported in fractions. MCSE measures sampling noise under chosen assumptions, not model error or real policy quality.

Ten-line private LP-style defence

1. This is a private synthetic model, not a real fund forecast or offer.
2. All four strategies conserve 40 M and include the same stated exclusions.
3. Company outcomes use chosen five-point probabilities, not fitted market data.
4. Observable scores are no-edge in baseline; future-winner selection is hindsight only.
5. Follow-on dollars pay twice the price, buying half the payoff per dollar.
6. Analytic and fixed-seed means differ;1,000 paths reduce sampling error only.
7. A deployed-only metric cannot be compared as if it used the same commitment denominator.
8. B leads expected gross multiples under these inputs, not under every plausible real policy.
9. Tail, correlated-shock and funded-cost scenarios materially change risk or results.
10. Change the choice only after testing actual score edge, prices, access, costs, timing, cash needs and downside tolerance.

Method context, not validation of toy probabilities:

https://avc.com/2017/01/reserves/
https://www.kauffmanfellows.org/journal/strategies-for-optimal-follow-on-investments
https://www.kauffmanfellows.org/journal/multi-stage-portfolio-construction-with-monte-carlo
https://angel.co/pdf/followon.pdf